Customer Shopping Behavior Analysis

Python · PostgreSQL · Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

Item	Detail
Rows	3,900
Columns	18
Customer demographics	Age, Gender, Location, Subscription Status
Purchase details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
Missing data	37 values in the Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and .describe() for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:** Created an age_group column by binning customer ages, and a purchase_frequency_days column from purchase data.
- **Data Consistency Check:** Verified whether discount_applied and promo_code_used were redundant; dropped promo_code_used.
- **Database Integration:** Connected the Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Summary statistics (numeric columns)

Statistic	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900	3900	3900	3863	3900
mean	1950.50	44.07	59.76	3.75	25.35
std	1125.98	15.21	23.69	0.72	14.45
min	1	18	20	2.5	1
25%	975.75	31	39	3.1	13
50%	1950.50	44	60	3.8	25
75%	2925.25	57	81	4.4	38
max	3900	70	100	5.0	50

Summary statistics (categorical columns)

Column	Unique values	Most frequent	Frequency
Gender	2	Male	2652
Item Purchased	25	Blouse	171
Category	4	Clothing	1737
Location	50	Montana	96
Size	4	M	1755
Color	25	Olive	177
Season	4	Spring	999

Column	Unique values	Most frequent	Frequency
Subscription Status	2	No	2847
Shipping Type	6	Free Shipping	675
Discount Applied	2	No	2223
Promo Code Used	2	No	2223
Payment Method	6	PayPal	677
Frequency of Purchases	7	Every 3 Months	584

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions.

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

gender	revenue
Female	75,191
Male	157,890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount. The query returned 839 rows; the first 10 are shown.

customer_id	purchase_amount
2	64
3	73
4	90
7	85

customer_id	purchase_amount
9	97
12	68
13	72
16	81
20	90
22	62

3. Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	Average purchase amount (USD)
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1,053	59.49	62,645.00
No	2,847	59.87	170,436.00

6. Discount-Dependent Products

Identified the 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	Number of Customers
Loyal	3,116
New	83
Returning	701

8. Top 3 Products per Category

Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171

item_rank	category	item_purchased	total_orders
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Checked whether customers with more than 5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2,518
Yes	958

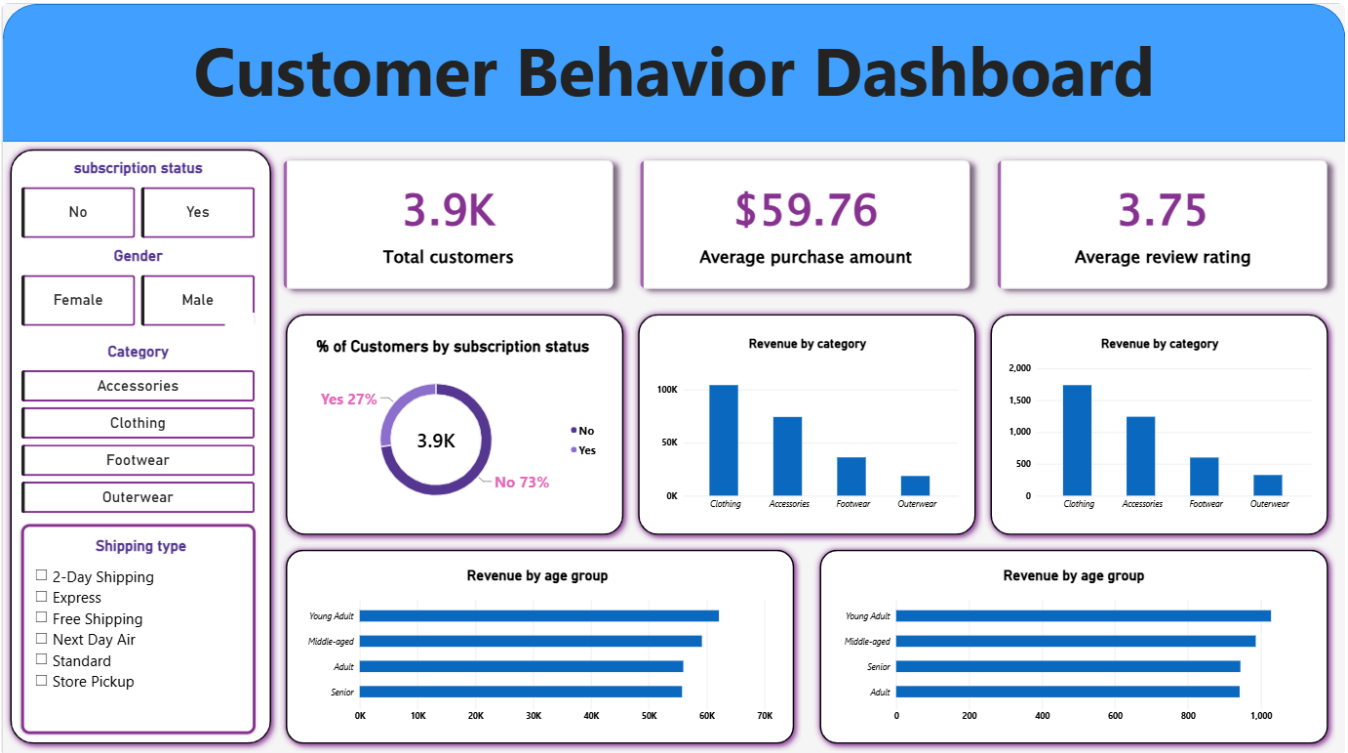
10. Revenue by Age Group

Calculated the total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually. It shows three headline KPIs (3.9K total customers, $59.76 average purchase amount, 3.75 average review rating), subscription share (Yes 27% / No 73%), revenue and sales by category, and revenue and sales by age group, with slicers for subscription status, gender, category and shipping type.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.

- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.